MODULE 5 AUTONOMIC NERVOUS SYSTEM

The Autonomic Nervous System.

THE AUTONOMIC NERVOUS SYSTEM

Covers the difference between the somatic and autonomic systems, the two-neuron motor pathway, the named autonomic ganglia, the sympathetic and parasympathetic divisions, and the autonomic and enteric plexuses. The focus is on the structures and the job each one does.

BY THE END

1. Contrast the somatic and autonomic motor systems.
2. Describe the two-neuron motor pathway and name the autonomic ganglia.
3. Compare the sympathetic and parasympathetic divisions and the plexuses.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: The autonomic nervous system

drsrennie-stack.github.io/new-build-bio4-solano/ans-concept-videos.html

The Autonomic Nervous System, an Overview

The autonomic nervous system is the involuntary motor division. It quietly regulates the organs, adjusting them without conscious effort or awareness.

WHAT THE SYSTEM IS AND WHAT IT CONTROLS

Structure	What it is
Autonomic nervous system	The involuntary division of the motor system; it controls cardiac muscle, smooth muscle, and glands
Autonomic effectors	The targets of the ANS: cardiac muscle, smooth muscle, and glands
Autonomic sensory neurons	Neurons that carry input from receptors in the viscera and blood vessels to the CNS
Autonomic motor neurons	Neurons that carry commands out to the visceral effectors, either exciting or inhibiting them
Two divisions	The sympathetic and the parasympathetic divisions, which usually act in opposite directions on the same organ
Involuntary control	Most autonomic activity cannot be consciously altered, although emotion and pain can influence it

Somatic Versus Autonomic

The somatic and autonomic systems are both motor, but they differ in what they control, how many neurons they use, and what effect they have. Compare them.

THE TWO MOTOR SYSTEMS SIDE BY SIDE

Feature	Somatic nervous system	Autonomic nervous system
Effector	Skeletal muscle	Cardiac muscle, smooth muscle, and glands
Control	Voluntary	Involuntary
Motor neurons in the pathway	One, from the CNS straight to the muscle	Two, with a synapse in a ganglion
Effect on the effector	Always excitation; it causes contraction	Either excitation or inhibition
Awareness	Normally conscious	Normally not conscious

The Two-Neuron Motor Pathway

Where the somatic system reaches its muscle with a single neuron, the autonomic system always uses two neurons in a row, with a synapse in a ganglion between them.

THE TWO NEURONS AND THE GANGLION BETWEEN THEM

Structure	What it is
Preganglionic neuron	The first neuron; its cell body sits in the brain or spinal cord, and its myelinated axon runs out to an autonomic ganglion
Autonomic ganglion	The cluster of cell bodies where the two neurons synapse
Postganglionic neuron	The second neuron; its cell body sits in the ganglion, and its unmyelinated axon runs from the ganglion to the visceral effector
The pathway	The preganglionic neuron synapses in the ganglion with the postganglionic neuron, which then reaches the effector

KEY POINT

The preganglionic axon is myelinated and the postganglionic axon is not. The synapse in the ganglion is the dividing line, and it is the one feature the somatic system does not have at all.

The Autonomic Ganglia

The autonomic ganglia fall into two groups by division. The sympathetic ganglia sit close to the spinal cord; the parasympathetic ganglia sit close to or within the organ.

Sympathetic ganglia

THE GANGLIA NEAR THE VERTEBRAL COLUMN

Structure	What it is
Sympathetic trunk ganglia	Also called paravertebral ganglia; a vertical chain on each side of the vertebral column, from the base of the skull to the coccyx
Prevertebral ganglia	Also called collateral ganglia; ganglia anterior to the vertebral column, near the large abdominal arteries
The five prevertebral ganglia	The celiac, superior mesenteric, inferior mesenteric, aorticorenal, and renal ganglia

Parasympathetic ganglia

THE GANGLIA NEAR THE TARGET ORGAN

Structure	What it is
Terminal ganglia	The parasympathetic ganglia, located near or within the wall of the target organ
The four named terminal ganglia of the head	The ciliary, pterygopalatine, submandibular, and otic ganglia

The Two Divisions

The two divisions are told apart first by where they leave the CNS, and then by the work they do.

The sympathetic division

FIGHT OR FLIGHT

Structure	What it is
Sympathetic division	The fight or flight division; also called the thoracolumbar division
Origin	Preganglionic cell bodies in the lateral gray horns of the T1 to L2 spinal segments
Role	It takes over during exercise, excitement, emergency, and embarrassment
Sample effects	Dilates the pupils, speeds the heart, opens the airways, and slows digestion

The parasympathetic division

REST AND DIGEST

Structure	What it is
Parasympathetic division	The rest and digest division; also called the craniosacral division
Origin	Preganglionic cell bodies in the nuclei of four cranial nerves (III, VII, IX, and X) and in the S2 to S4 spinal segments
Role	It conserves energy and runs routine housekeeping: digestion, defecation, and urination
Sample effects	Constricts the pupils, slows the heart, constricts the airways, and stimulates digestion

Comparing the Two Divisions

Side by side, the two divisions differ in their origin in the CNS, the location of their ganglia, and their overall job. Compare them.

SYMPATHETIC AGAINST PARASYMPATHETIC

Feature	Sympathetic	Parasympathetic
Other name	Thoracolumbar division	Craniosacral division
Origin in the CNS	The T1 to L2 spinal segments	The brainstem and the S2 to S4 spinal segments
Location of ganglia	Close to the spinal cord	Close to or within the effector organ
Overall role	Fight or flight; prepares the body for action	Rest and digest; conserves energy

KEY POINT

The two other names are the whole map of the origins. Thoracolumbar means the fibers leave from T1 to L2, and craniosacral means they leave from the brainstem and S2 to S4. The ganglia then follow: sympathetic near the cord, parasympathetic at the organ.

The Autonomic and Enteric Plexuses

Autonomic axons and ganglia travel together in networks called plexuses. A separate set of networks, the enteric plexuses, runs the wall of the gut.

Autonomic plexuses

THE NAMED NETWORKS OF THE TRUNK

Structure	What it is
Autonomic plexus	A network of autonomic axons and ganglia in the thorax, abdomen, and pelvis
Cardiac plexus	Supplies the heart
Pulmonary plexus	Supplies the bronchial tree
Celiac (solar) plexus	The largest; it surrounds the celiac trunk and supplies most of the abdominal organs
Superior and inferior mesenteric plexuses	Supply the small and large intestine
Hypogastric plexus	Supplies the pelvic organs
Renal plexus	Supplies the kidneys and ureters

Enteric plexuses

THE NETWORKS IN THE WALL OF THE GUT

Structure	What it is
Enteric plexuses	The neural networks within the wall of the GI tract, the structural core of the enteric nervous system
Myenteric plexus	Lies between the longitudinal and circular muscle layers; it controls the movement of the gut
Submucosal plexus	Lies between the circular muscle layer and the muscularis mucosae; it controls the secretions of the gut

Tracing an Autonomic Pathway

Every autonomic motor command travels the same four-step route from the central nervous system to the organ.

1. **Preganglionic neuron** the cell body in the brain or cord sends its axon out toward a ganglion
2. **Autonomic ganglion** the preganglionic axon synapses here with the postganglionic neuron
3. **Postganglionic neuron** its axon carries the signal the rest of the way out
4. **Effector tissue** the cardiac muscle, smooth muscle, or gland receives the command

TWO DIVISIONS, OPPOSITE EFFECTS

Because the two divisions usually pull an organ in opposite directions, the body can fine-tune an organ by easing off one division rather than only by pushing the other. Heart rate, for example, rises when **parasympathetic** input is withdrawn, even before the sympathetic division speeds things up.

The **vagus nerve**, CN X, carries most of the parasympathetic supply to the chest and abdomen. Its long reach is why a single nerve can slow the heart, ease breathing, and drive digestion all at once.

The **enteric nervous system** is independent enough to be called the brain of the gut. Smooth muscle in the GI tract keeps contracting in a rhythm and glands keep secreting even when the link to the CNS is cut, which is why a denervated bowel can still move food along.